

VIDEOVERITAS: AI-Generated Video Detection via Perception Pretext Reinforcement Learning

Hao Tan^{1 2 3} Jun Lan^{† 3} Senyuan Shi² Zichang Tan⁴ Zijian Yu³ Huijia Zhu³ Weiqiang Wang³
Jun Wan^{1 2 5} Zhen Lei^{1 2 5}

Project: <https://github.com/EricTan7/VideoVeritas>

Abstract

The growing capability of video generation poses escalating security risks, making reliable detection increasingly essential. In this paper, we introduce **VIDEOVERITAS**, a framework that integrates fine-grained perception and fact-based reasoning. We observe that while current multi-modal large language models (MLLMs) exhibit strong reasoning capacity, their granular perception ability remains limited. To mitigate this, we introduce *Joint Preference Alignment* and *Perception Pretext Reinforcement Learning (PPRL)*. Specifically, rather than directly optimizing for detection task, we adopt general spatiotemporal grounding and self-supervised object counting in the RL stage, enhancing detection performance with simple *perception pretext tasks*. To facilitate robust evaluation, we further introduce **MintVid**, a light yet high-quality dataset containing 3K videos from 9 state-of-the-art generators, along with a real-world collected subset that has factual errors in content. Experimental results demonstrate that existing methods tend to bias towards either *superficial* reasoning or *mechanical* analysis, while **VIDEOVERITAS** achieves more balanced performance across diverse benchmarks.

1. Introduction

With the rapid advancement of video generation, our digital lives have been immensely enriched. AI-generated videos

[†]Project Lead ¹School of Advanced Interdisciplinary Sciences, UCAS ²MAIS, Institute of Automation, Chinese Academy of Sciences ³Ant Group ⁴Shenzhen Institute of Advanced Technology (SIAT), Chinese Academy of Sciences ⁵School of Artificial Intelligence, UCAS. Correspondence to: Jun Wan <jun.wan@ia.ac.cn>.

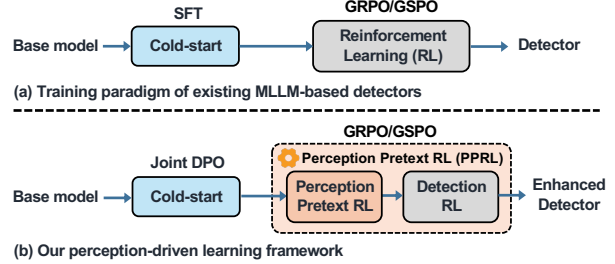


Figure 1. **Comparison with previous training pipeline.** (a) Existing MLLM-based detectors typically adopt supervised fine-tuning (SFT) or reinforcement learning (RL) on detection task. (b) Our framework adopts Joint DPO for cold-start, and further enhances the detection capacity by introducing simple perception pretext tasks in the RL stage.

have become pervasive on short-video platforms such as TikTok, which, while providing unprecedented entertainment, raise significant concerns regarding security. AI-generated video detection, which aims at determining the authenticity of a video, has therefore emerged as a critical topic.

In this landscape, various methods (Chen et al., 2024; Ma et al., 2025; Zheng et al., 2025b; Internò et al., 2025; Zhang et al., 2025b; Corvi et al., 2025) and datasets (Chen et al., 2024; Ni et al., 2025; Wen et al., 2025a;b; Fu et al., 2025b; Li et al., 2025a) have been proposed. Beyond binary discrimination, recent efforts (Wen et al., 2025b; Park et al., 2025; Fu et al., 2025b; Li et al., 2025a; Gao et al., 2025) focus on providing artifacts explanations. However, current methods face **two challenges**: (1) Even state-of-the-art (SoTA) multi-modal large language models (MLLMs) struggle to capture human-perceivable artifacts (Wang et al., 2025b). To mitigate this, BusterX++ (Wen et al., 2025b) adopts pure Reinforcement Learning (RL) on large-scale detection dataset, but the resulting model falls into superficial analysis, focusing on coarse features like environment and lighting. (2) The task-oriented fine-tuning leads to *mechanical grounded analysis*. Methods like Skyra (Li et al., 2025a) and DeepTraceReward (Fu et al., 2025b) construct grounded Chain-of-Thought (CoT) for Supervised Fine-Tuning (SFT), aiming to inject fine-grained perceptual capacities to base

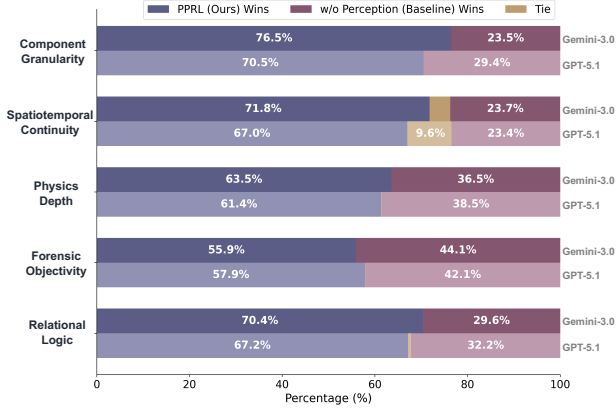


Figure 2. To understand why PPRL enhances detection, we characterize the model’s reasoning behavior across five distinct dimensions, finding that PPRL effectively shapes better reasoning behavior. For instance, model trained with PPRL tends to break down a whole scene into specific objects (i.e., “Component Granularity”: 76.5% win rate). Details are provided in Sec. 4.4.

model. However, the resulting model fails to conduct basic fact-based reasoning on AI parodies.

To mitigate these issues, we introduce **VIDEOVERITAS**, an MLLM-based detector that integrates fine-grained perception and general reasoning. As shown in Figure 1, our approach leverages a two-stage training pipeline. **In the first stage**, rather than performing SFT on large-scale database, we introduce Joint Preference Alignment. Specifically, we construct question-answering (QA) reports to select data with diversified artifacts. To integrate perceptual and reasoning abilities, we incorporate both response-level and video-level alignments, leveraging the base model itself as a strong guidance (e.g., using “anti-label” strategy to probe the intrinsic hallucinations as non-preference). **In the second stage**, we introduce Perception Pretext RL (PPRL). Rather than directly optimizing for AIGC detection task, we adopt (1) general spatiotemporal grounding and (2) self-supervised object counting as perception pretext tasks, which yield promising improvements on detection without requiring label-intensive annotations. As shown in Figure 2, we also empirically demonstrate that PPRL leads to *better reasoning behavior*, which therefore *benefits* detection.

Moreover, videos in existing datasets (e.g., GenVideo (Chen et al., 2024) and GenVidBench (Ni et al., 2025)) primarily come from early models like ModelScope, which suffer from suboptimal temporal consistency. To support more robust evaluation, we present **MintVid**, a light AI-generated video dataset spanning three parts: (1) 1.5K highly realistic videos from 6 powerful proprietary models, each including both T2V and TI2V, (2) 2K deepfake videos using 3 SoTA public models, and (3) a fact-based subset collected from short-video platforms, which could be identified through factual reasoning. Together with available benchmarks, e.g., GenBuster++ (Wen et al., 2025b), the models are evaluated

on a holistic and challenging setting.

Building on MintVid, we reveal that binary detectors like DeMamba (Chen et al., 2024) and ReStraV (Internò et al., 2025) struggle to yield satisfactory results. MLLM with cold-start (e.g., Skyra) exhibits notably low recall on fact-based subset, while pure RL (e.g., BusterX++) demonstrates degraded results on those challenging subsets. In contrast, **VIDEOVERITAS** achieves a more balanced performance.

To sum up, our main contributions are:

- We introduce Perception Pretext RL **algorithm**, which leverages simple perception pretext tasks to elevate detection performance. This method can be seamlessly integrated into existing RL-paradigm framework.
- We establish **VIDEOVERITAS**, a **framework** that integrates fine-grained perception and fact-based reasoning for AI-generated video detection, achieving superior results over previous methods across multiple datasets.
- We introduce MintVid, a light yet high-quality **dataset** containing 3K videos from 9 SoTA generators. MintVid facilitates evaluations on three aspects, including general-content, facial and fact-based scenarios.

2. Related Work

2.1. AI-Generated Video Detection and Datasets

Early studies primarily focused on image-level forgery detection (Ojha et al., 2023; Yan et al., 2024b; Tan et al., 2024; Nguyen et al., 2024; Fu et al., 2025c; Yan et al., 2024a;c; Yang et al., 2025b). With the increasing capabilities of video generation, various efforts are made for video detection (Chen et al., 2024; Bai et al., 2024; Liu et al., 2024b; Ma et al., 2025; Zheng et al., 2025b; Internò et al., 2025; Zhang et al., 2025b; Corvi et al., 2025). For instance, D3 (Zheng et al., 2025b) revealed the discrepancy in the second-order differences of features between real and AI-generated videos, enabling a training-free detection through simple pre-trained encoders. Sharing similar insights, ReStraV (Internò et al., 2025) leverages the statistical discrepancy between pre-trained representations for discrimination. NSG-VD (Zhang et al., 2025b) introduces a novel perspective based on physical conservation principles. Similar to recent “bias-free” approaches (Zhou et al., 2024; Yan et al., 2024b; Kashiani et al., 2025; Chen et al., 2025) in image domain, WaveRep (Corvi et al., 2025) explores unbiased training using video VAE and frequency-level alignment, greatly enhancing detection robustness under H.264 compression. Besides, some methods also target deepfake video detection (Haliassos et al., 2021; Wang et al., 2023; Xu et al., 2023; Han et al., 2025; Kim et al., 2025; Nguyen et al., 2025; Yan et al., 2025), incorporating facial

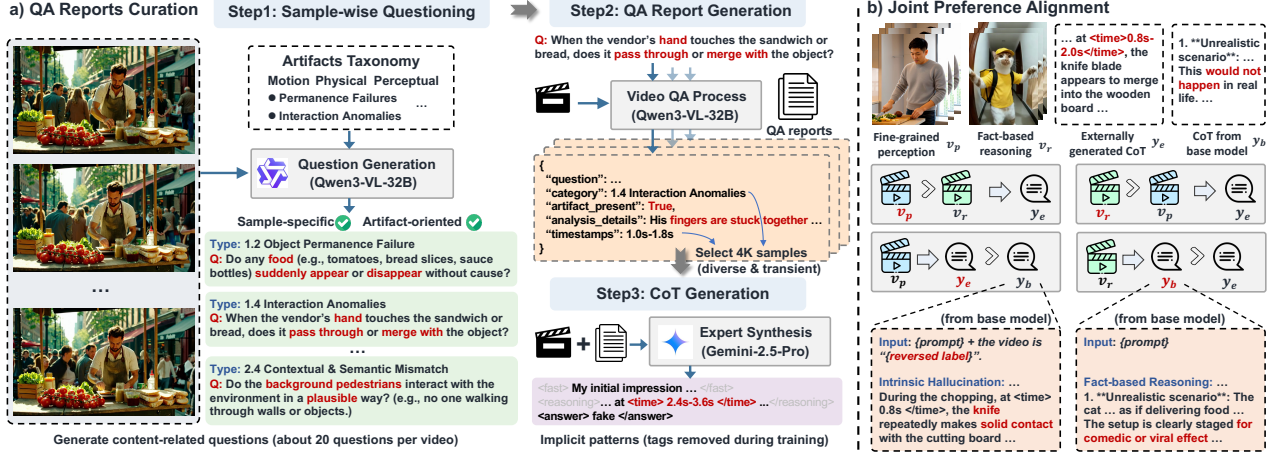


Figure 3. Overview of our Joint Preference Alignment stage. (a) We generate Question-Answering (QA) Reports to select diverse data and curate high-quality Chain-of-Thought (CoT). It involves generating artifact-oriented questions and creating detailed QA reports. (b) Joint DPO constructs preference pairs for both response-level and video-level alignments, leveraging external CoT and the base model’s own reasoning to effectively guide the model. The artifacts taxonomy is provided in Appendix B.

priors to achieve robust detection. However, most methods are evaluated on previous datasets, e.g., GenVideo (Chen et al., 2024) and GenVidBench (Ni et al., 2025), where most videos are derived from *outdated* generative models, suffering from limited temporal consistency. Although various benchmarks (Du et al., 2025; Wang et al., 2025a; Huang et al., 2025; Li et al., 2025b) have been proposed recently, they primarily focus on image domain. In this paper, we introduce a video dataset to facilitate more robust evaluation.

2.2. Multimodal Large Language Models

Brief Review of Generic MLLMs. To encode the temporal information, Qwen2-VL (Wang et al., 2024) employs Multimodal Rotary Position Embedding (M-RoPE). Qwen2.5-VL (Bai et al., 2025b) further incorporates absolute temporal encoding, while Qwen3-VL (Bai et al., 2025a) introduces explicit timestamp tokens, achieving strong performance on generic video benchmarks like VideoMME (Fu et al., 2025a). To adaptively encode long videos, Key-VL1.5 (Yang et al., 2025a) proposes a slow-fast frames encoding strategy, while VideoL-LaMA3 (Zhang et al., 2025a) introduces a Differential Frame Pruner, greatly improving performance on long video understanding. Although their semantic-level understanding capabilities have greatly advanced, their fine-grained spatiotemporal perception remains limited (Zhao et al., 2025).

MLLMs for AI-Generated Video Detection. Leveraging the semantic understanding capabilities of MLLMs, several studies have explored explainable AI-generated video detection (Song et al., 2024; Jiang et al., 2025; Xu et al., 2025; Gao et al., 2025; Fu et al., 2025b; Park et al., 2025; Wen et al., 2025a;b; Li et al., 2025a). DAVID-XR1 (Gao

et al., 2025) and VidGuard-R1 (Park et al., 2025) employ Supervised Fine-Tuning (SFT) using Chain-of-Thought (CoT) generated by Gemini-2.5-Pro and Qwen2.5-VL-72B, respectively. DeepttraceReward (Fu et al., 2025b) introduces “human-perceivable” explainability and involves timestamps and bounding boxes in the reasoning process, while Skyra (Li et al., 2025a) uses 4K manually annotated samples to inject the grounding capacities into base model. In contrast, BusterX++ (Wen et al., 2025b) supposes that low-quality cold starts can affect the generic reasoning ability and therefore adopts pure Reinforcement Learning (RL) on large-scale training data. In this paper, we introduce Perception Pretext RL, which achieves promising gains without the burden of human-annotated AIGC data.

3. Method

In this section, we present a detailed description of the two-stage training pipeline and MintVid dataset.

3.1. Joint Preference Alignment

QA reports generation and CoT curation. As shown in Figure 3, for each video, we first generate various *artifact-oriented* questions that are strongly *related* to its content. Then we use Qwen3-VL-32B (Bai et al., 2025a) to answer each question, forming a comprehensive QA report for each video. Each item contains the type of artifacts and exact timestamps. Such strategy offers two benefits: (1) It enables *independent* and label-agnostic perception of specific artifact, showing fewer hallucinations than directly prompted with Ground-Truth labels. (2) It provides a valuable reference for data sampling. Based on this pipeline, we fil-

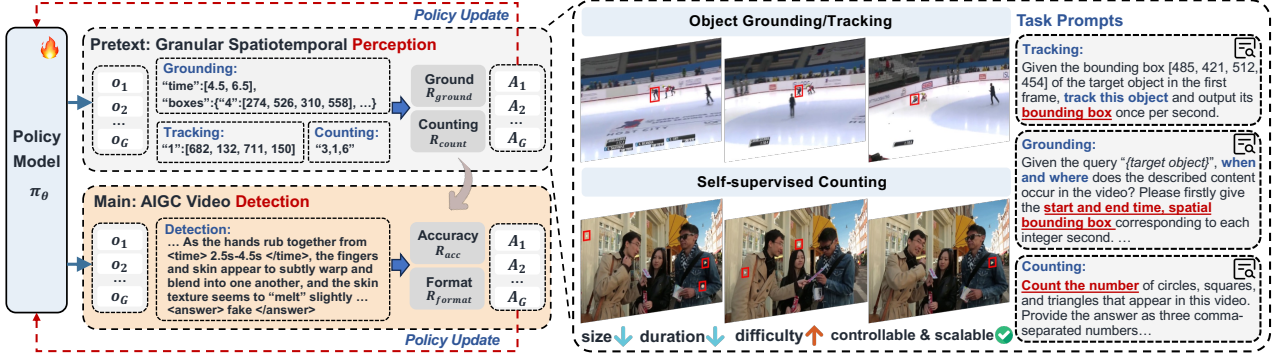


Figure 4. **Perception Pretext RL (PPRL)**. **Left**: Perception is taken as a foundational phase to detection. The pretext phase can be implemented with various perception-oriented tasks, e.g., spatiotemporal grounding and object counting. **Right**: Examples of perception pretext tasks. Grounding and tracking data is sampled from OneThinker (Feng et al., 2025). The model is prompted to output exact bounding boxes and timestamps. Self-supervised counting is controllable by adjusting the size and duration of the objects, and the model is required to output the exact quantity of each shape.

ter 4K samples from the database comprising GenBuster-200K (Wen et al., 2025a), RewardData (Fu et al., 2025b), TalkingheadBench (Xiong et al., 2025) and several fact-based videos, where v_p and v_r denote videos that require fine-grained perception and fact-based reasoning, respectively. We utilize Gemini-2.5-Pro to generate CoT based on its own reasoning and QA reports. Following Veritas (Tan et al., 2025), we adopt a structured reasoning design, but remove the explicit “tags” during training, encouraging the model to learn the underlying thinking pattern inherently.

Response-level alignment. Suppose previous annotated CoT is denoted as y_e , we make full use of the base model itself for more effective alignments: (1) We utilize “anti-label” strategy (i.e., insert the reversed label in the prompt) to trigger the base model’s *intrinsic* hallucinations, designating these as non-preferred responses. (2) For videos requiring factual reasoning, the base model’s outputs are directly leveraged as preferred responses. Let y_b denote the CoT generated from base model. For v_p we assume that y_e is preferred over y_b (i.e., $y_e \succ y_b$), whereas for v_r , we suppose the opposite preference (i.e., $y_b \succ y_e$). Given the input query x , the response-level preference objective is formulated as:

$$u_t(x, v, y_w, y_l) = \beta \log \frac{\pi_\theta(y_w | v, x)}{\pi_{\text{ref}}(y_w | v, x)} - \beta \log \frac{\pi_\theta(y_l | v, x)}{\pi_{\text{ref}}(y_l | v, x)}, \quad (1)$$

$$\mathcal{L}_{\text{DPO}_t} = -\mathbb{E}_{(x, v_p, y_e, y_b)} [\log \sigma(u_t(x, v_p, y_e, y_b))] - \mathbb{E}_{(x, v_r, y_b, y_e)} [\log \sigma(u_t(x, v_r, y_b, y_e))],$$

where $\sigma(\cdot)$ denotes sigmoid function.

Video-level alignments. To encourage the model to generate preferred outputs based on pure visual information, we involve video-level alignments. Specifically, we encourage the model to output fine-grained analysis y_e based on video v_p (i.e., $v_p \succ v_r$), while outputting factual reasoning y_b when given video v_r (i.e., $v_r \succ v_p$). The video-level

preference objective is computed as:

$$u_v(x, y, v_w, v_l) = \beta \log \frac{\pi_\theta(y | v_w, x)}{\pi_{\text{ref}}(y | v_w, x)} - \beta \log \frac{\pi_\theta(y | v_l, x)}{\pi_{\text{ref}}(y | v_l, x)}, \quad (2)$$

$$\mathcal{L}_{\text{DPO}_v} = -\mathbb{E}_{(x, y_e, v_p, v_r)} [\log \sigma(u_v(x, y_e, v_p, v_r))] - \mathbb{E}_{(x, y_b, v_r, v_p)} [\log \sigma(u_v(x, y_b, v_r, v_p))].$$

The final objective for the Joint DPO is defined as:

$$\mathcal{L}_{\text{J-DPO}} = \mathcal{L}_{\text{DPO}_t} + \mathcal{L}_{\text{DPO}_v}. \quad (3)$$

3.2. Perception Pretext RL (PPRL)

As illustrated in Figure 4, rather than directly optimizing for AIGC detection task, we introduce perception pretext tasks. By first encouraging the model to accurately perceive subtle and transient targets, the fine-grained perception capacity is sharpened, benefiting the following detection task. Specifically, we introduce two types of perception tasks: (1) general spatiotemporal grounding and (2) self-supervised object counting.

General grounding. This includes spatiotemporal grounding and object tracking. For tracking task, given the bounding box B_1 in the first frame, the model is required to predict a sequence of bounding boxes $\{\hat{B}_i\}_{i=2}^N$ to track the object across the video. Suppose $\{B_i\}_{i=2}^N$ denotes the ground-truth boxes, the reward is measured as the mean Intersection over Union (IoU) over all frames:

$$R_{\text{track}} = \frac{1}{N-1} \sum_{i=2}^N \frac{|\hat{B}_i \cap B_i|}{|\hat{B}_i \cup B_i|}. \quad (4)$$

For spatiotemporal grounding, the model is required to predict both the temporal span t of a query and its bounding boxes $\{\hat{B}_i\}_{i=1}^M$ across the video. Given the ground-truth t_y and $\{B_i\}_{i=1}^M$, the reward is measured according to both the

timestamp and spatial boxes:

$$R_{\text{ground}} = \frac{1}{2} \frac{|\hat{t} \cap t_y|}{|\hat{t} \cup t_y|} + \frac{1}{2} \sum_{i=1}^M \frac{|\hat{B}_i \cap B_i|}{|\hat{B}_i \cup B_i|}. \quad (5)$$

There are abundant samples that are publicly available. In our case, all data are sampled from OneThinker (Feng et al., 2025), with easy samples filtered out, retaining only small and short-duration targets for training.

Self-supervised object counting. For a given video, we randomly add a set of objects with randomized attributes (e.g., size, duration and position) to the frames. The model is required to output the counts for each shape category in the video, including circles, squares and triangles. For each shape s , the reward $R_{\text{count},s}$ is inversely proportional to the relative prediction error:

$$R_{\text{count},s} = \max(0, 1 - \frac{|\hat{y}_s - y_s|}{y_s + \epsilon}), \quad (6)$$

where $\hat{y}_s$ and y_s denotes the predicted and ground-truth counts, respectively. ϵ is a small constant (e.g., 10^{-6}) to ensure numerical stability. The final counting reward R_{count} is averaged from all shapes:

$$R_{\text{count}} = \frac{1}{S} \sum_{s=1}^S R_{\text{count},s}. \quad (7)$$

Since the object size and duration are fully controllable, we can generate diverse and challenging training signals at scale. In our case, we randomly select videos from OpenVid-1M (Nan et al., 2024) to construct learning samples. Only with precise spatiotemporal perception can the model correctly count different shapes. This enables efficient scaling without requiring any manual annotations.

AIGC detection. For the detection task, we utilize accuracy reward R_{acc} and format reward R_{format} , where $R_{\text{acc}} = 1$ when the answer is correct and $R_{\text{acc}} = 0$ otherwise. R_{format} is given when the answer is enclosed in `<answer>` `</answer>` tags. The reward for detection task is:

$$R_{\text{detection}} = R_{\text{acc}} + \alpha R_{\text{format}}, \quad (8)$$

where $\alpha = 0.2$ is to control the effect of format reward. Specifically, we use 3K samples for general grounding, 2K samples for self-supervised object counting, and 10K samples (from database in Sec. 3.1) for AIGC detection. The perception task and detection task are performed in a sequential manner. Since our method is suitable for most R1-style algorithms (Liu et al., 2024a), we adopt Group Sequence Policy Optimization (GSPO) (Zheng et al., 2025a) for training. Instead of relying on manually annotated AIGC data or elaborate reward design, we demonstrate that simple RL on perception pretext tasks could achieve promising gains.

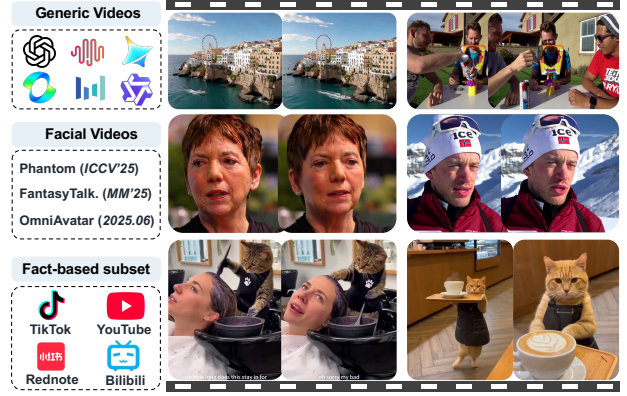


Figure 5. Examples of the MintVid dataset. MintVid comprises three different parts, which facilitates more complete evaluation.

3.3. MintVid Dataset

As shown in Figure 5, MintVid comprises three parts:

General content videos. 1.5K videos from 6 powerful proprietary models: Jimeng 3.0 Pro, Seedance 1.0 Pro, Kling 2.5-Turbo, Wan2.5, MiniMax Hailuo 2.3, and Sora2. For each model, we sample 100 meticulous prompts from VideoVerse (Wang et al., 2025d), PhyGenBench (Meng et al., 2024) and VMBench (Ling et al., 2025), and 200 prompts from OpenVid-1M (Nan et al., 2024) captions. Among them, 100 prompts are paired with the corresponding first frame to generate TI2V data. After manual filtering, 1.5K videos are kept. Real videos are sampled from OpenVid-1M.

Facial videos. 2K videos from 3 SoTA human-centric video generative models: OmniAvatar (Gan et al., 2025), FantasyTalking (Wang et al., 2025c), and Phantom (Liu et al., 2025), all using their 14B variants. Real videos are sampled from VFHQ (Xie et al., 2022) and HDTF (Zhang et al., 2021). We curate input prompts to yield high-quality videos. Please see Appendix A for more details.

Fact-based subset. This part is often overlooked in existing datasets. For videos that violate objective facts, general MLLMs can perform correct reasoning, whereas fine-tuned models may fail due to formulaic analysis. To this end, we collect over 200 videos (including both real and fake) from TikTok, YouTube, and Bilibili, and conduct manual filtering to ensure all videos can be verified using objective facts.

4. Experiments

4.1. Experimental Setup

Evaluation Protocols. In-domain (ID) testing includes test sets of GenBuster-200K (Wen et al., 2025a) and Talking-headBench (Xiong et al., 2025). Out-of-domain (OOD) testing involves existing benchmarks GenBuster++ (Wen et al., 2025b), LOKI (Ye et al., 2024) (using their video set), and

Table 1. Performance comparison on video datasets. Accuracy is reported except for D3 (Zheng et al., 2025b), where * means Average Precision (AP) is adopted following the official guideline. The final average performance is calculated by first averaging the results across ID (in-domain), OOD (out-of-domain) and OOD-MintVid, and then taking the average of the three values. The best results are **bolded** and the second best are underlined. More metrics are provided in Appendix D.

Method	ID	OOD						OOD-MintVid											Avg.
		GenBuster++	LOKI	Vidu Q1	Gen-4	Veo3	Emu3	Phantom-14B	OmniAvatar	FantasyTalking	Seedance1.0Pro	Jimeng3.0Pro	Kling2.5-Turbo	Hailuo2.3	Wan2.5	Sora2	Fact		
Small Vision Models																			
DeMamba	87.6	86.8	71.1	80.7	83.4	79.4	85.9	57.4	55.2	62.2	49.2	46.2	46.6	41.1	49.3	41.4	73.3	73.7	
D3* (ICCV'25)	49.5	46.8	39.1	55.1	89.0	66.1	90.0	54.8	36.4	44.1	61.6	48.7	51.7	78.9	50.5	54.6	32.4	55.1	
NSG-VD (NIPS'25)	52.4	53.3	55.1	53.4	58.2	64.3	50.7	54.3	53.8	58.5	53.8	58.0	56.0	69.4	57.0	57.8	52.8	55.1	
ReStraV (NIPS'25)	50.7	48.9	64.8	54.8	52.8	59.8	95.2	49.6	49.3	49.9	36.9	48.4	47.2	30.2	40.9	45.4	65.0	53.2	
Generic MLLMs																			
Qwen2.5-VL-7B	54.1	59.5	49.4	58.8	53.3	60.3	99.2	50.7	50.0	49.7	63.6	49.4	51.6	53.1	55.8	50.4	87.1	57.9	
Qwen3-VL-8B	65.1	62.4	56.6	60.2	59.4	62.9	99.9	53.3	50.6	50.3	64.2	51.8	55.1	54.1	66.7	50.8	91.9	63.6	
InternVL3.5-8B	51.7	59.6	44.1	56.1	53.4	56.7	99.7	51.3	50.2	49.5	62.6	49.8	49.1	50.2	53.9	54.1	91.7	56.5	
MiMo-VL-7B	65.8	54.3	53.6	65.5	65.8	66.0	98.8	58.4	50.1	50.6	69.7	56.7	61.7	56.1	66.0	55.4	89.9	64.8	
Keye-VL-1.5-8B	52.8	62.4	60.4	58.3	56.0	62.4	94.7	51.3	51.1	53.3	59.7	50.5	51.0	52.8	52.9	51.8	80.4	58.0	
GLM-4.5V	61.7	59.4	49.4	56.5	56.6	62.4	97.1	57.9	52.2	50.5	63.2	48.9	49.2	49.8	58.3	47.9	88.2	60.6	
Qwen3-VL-235B-A22B	66.7	64.6	57.9	57.4	65.7	63.4	98.5	54.1	50.9	51.2	65.8	49.1	32.3	49.8	63.5	50.5	93.6	63.6	
Gemini-2.5-Pro	80.5	71.8	70.2	70.8	76.0	70.4	93.3	57.8	55.8	55.6	69.4	60.2	65.6	62.7	65.7	60.9	88.3	73.4	
Gemini-3-Pro-Preview	79.4	82.5	74.9	83.1	76.3	85.7	89.3	65.5	65.3	61.5	81.4	72.1	73.1	74.3	73.1	63.5	91.4	77.8	
MLLM-based Video Forgery Detectors																			
BusterX++	77.1	79.6	70.9	53.0	89.9	76.2	98.9	62.3	48.6	47.6	71.8	71.7	82.3	70.4	87.3	58.7	88.8	74.7	
Skyra-RL	52.1	55.7	37.7	52.9	59.1	55.9	51.5	50.1	49.9	49.9	63.4	51.7	53.5	51.7	59.6	50.6	51.9	52.5	
VIDEOVERITAS (ours)	93.1	91.4	78.1	93.6	96.7	94.5	99.4	79.0	56.2	84.5	86.6	80.8	86.0	85.8	86.3	67.6	96.1	88.8	

high-quality subsets (e.g., Gen-4) from VBench-2 (Zheng et al., 2025c) and MVAD (Hu et al., 2025). MintVid is adopted as a challenging OOD testing. 4 SoTA binary detectors are trained on our dataset using their original preprocessing. Both public and proprietary MLLMs are evaluated using CoT prompting. We also evaluate 2 recent MLLM-based video forgery detectors, including BusterX++ (Wen et al., 2025b) and Skyra-RL (Li et al., 2025a).

Implementation Details. We implement the VIDEOVERITAS with Qwen3-VL-8B (Bai et al., 2025a). For Joint DPO, we train the model for 3 epochs using LoRA (Hu et al., 2022) (rank=64, $\alpha=128$), with a learning rate of 5×10^{-5} and batch size of 64. For PPRL, the model is trained for 2 epochs with the same LoRA setting, and the learning rate is set to 1×10^{-5} with a batch size of 32. The temperature is set to 1.0. Each epoch contains both perception and detection tasks. The videos are sampled at 3 FPS for up to 5 seconds, keeping their native resolutions.

4.2. Main Results

Comparisons to binary detectors. As shown in Table 1, recent statistical-based methods (Zheng et al., 2025b; Internò et al., 2025) are effective on certain subsets (e.g., 90.0% and 95.2% on Emu3), but their performance collapses to near-chance levels on most contemporary benchmarks, and even fails on in-domain subsets. This suggests that statistical discrepancies may not be a robust criterion for more realistic videos. DeMamba (Chen et al., 2024) performs well on previous datasets like GenBuster++, but suffers a sharp

performance drop on the more challenging MintVid dataset. In contrast, VIDEOVERITAS exhibits certain advantages, achieving 15.1% averaged gains over the previous best.

Comparisons to SOTA MLLMs. As shown in Table 1, although generic MLLMs demonstrate strong performance on the fact-based subset, their performance on other datasets remains limited, exhibiting low recall (presented in Table 10). Notably, Gemini-3.0-Pro-Preview outperforms most open-source models and exhibits a *clear lead* on MintVid. Its average performance even surpasses finetuned models like DeMamba and BusterX++, indicating that a *substantial gap still exists* between public and proprietary models. Moreover, our approach achieves an average improvement of 25.2% over base model, and it also outperforms those MLLMs with similar parameter scale, demonstrating the effectiveness of our training strategy.

Comparisons to MLLM-based detectors. As shown in Table 1, BusterX++ achieves 74.7% average accuracy, but shows suboptimal performance on MintVid, especially the facial subsets. Skyra-RL proves largely ineffective across most datasets. We suppose this may stem from the strict preprocessing applied to its training and testing data (e.g., 256p resizing), which could bias the model toward exploiting minor inconsistencies. In contrast, VIDEOVERITAS outperforms them on (1) existing datasets like GenBuster++ (+11.8%) and LOKI (+7.2%), (2) perception-heavy datasets like Jimeng3.0-Pro (+9.1%), and (3) fact-dependent subset, achieving a more balanced and robust detection capability.

Table 2. Ablations on the type of perception pretext tasks. 5K perception data are taken as default setting.

Pretext Tasks			ID		OOD		MintVid	
SSL	G-G	A-G	Acc	F1	Acc	F1	Acc	F1
×	×	×	92.9	93.4	91.2	92.1	77.8	73.3
✓	×	×	91.4	92.1	91.3	92.3	80.4	78.0
×	✓	×	92.3	92.8	91.1	92.2	79.8	76.7
×	×	✓	92.7	93.2	91.2	92.4	79.5	76.9
✓	✓	×	93.1	93.5	92.3	93.4	80.9	77.3
✓	×	✓	92.9	93.3	92.0	92.9	79.8	77.0
×	✓	✓	93.5	93.9	91.3	92.2	79.4	76.0
✓	✓	✓	92.7	93.2	91.3	92.2	78.6	75.0

Table 3. Ablations on the training stages. † means using format reward (Li et al., 2025a) to incentivize grounded reasoning.

Method	ID	OOD	MintVid	
			Others	Fact
Base	65.1	66.9	55.2	91.9
+ SFT	91.9	82.5	68.3	90.7
+ J-DPO	91.4	83.0	69.1	94.6
– DPO _v	91.3	82.7	69.1	92.9
– DPO _t	89.2	80.9	66.3	88.3
+ Pure RL	94.2	88.3	69.1	96.1
+ Pure RL [†]	94.5	90.2	70.9	87.5
+ J-DPO + RL	93.1	92.3	79.2	96.1

4.3. Ablation Studies

Ablations on the training stages. As shown in Table 3, SFT and Joint DPO (J-DPO) is trained on 4K samples. “J-DPO+RL” is further trained on 10K AIGC data and 5K perception data. Pure RL is trained on 20K samples, which ensures similar data scale to investigate the effect of cold-start. Specifically, we draw the following observations: (1) Supervised Fine-Tuning (SFT) achieves great improvements (e.g., +13.1% on MintVid-Others) but degrades the performance on Fact subset. In contrast, J-DPO effectively integrates the overall capacities, boosting accuracy on Fact subset to 94.6%. (2) Video-level alignment (DPO_v) is beneficial to synergizing the fact-based reasoning ability, while response-level alignment (DPO_t) is critical to overall performance. (3) The performance of pure RL on challenging subsets remains limited (e.g., 69.1% vs. 79.2% compared to ours), indicating the necessity of cold-start.

Ablations on PPRL: the type of perception tasks. Generally, the perception tasks can be categorized into three types with increasing annotation cost: (1) **Self-Supervised Learning (SSL)**, where we use object counting as a zero-cost pretext for foundational perception. (2) **Generic Grounding (G-G)**, where abundant public data can be utilized. (3) **Artifact Grounding (A-G)**, which is the most specialized and costly task. As shown in Table 2, each perception task independently improves the performance. Combining SSL and G-G yields further enhancements, achieving substantial gains on MintVid dataset as shown in Figure 6, suggesting that the *perceptual acuity gained from these tasks* provides

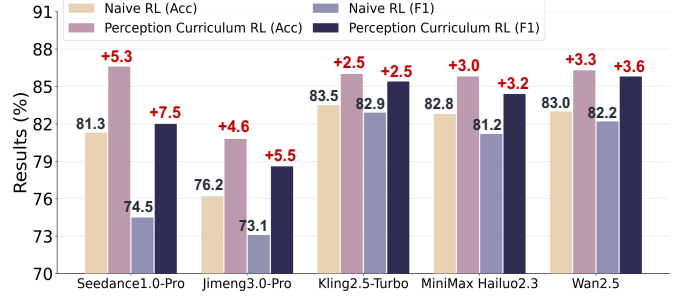


Figure 6. Improvements of the proposed PPRL. “SSL+G-G” is adopted in our method.

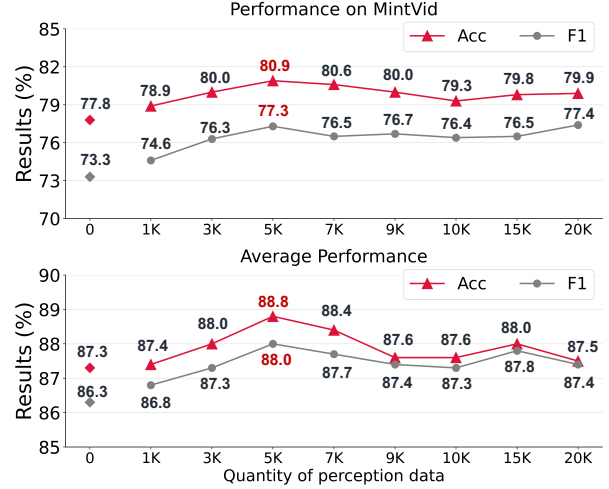


Figure 7. Ablations on the quantity of perception data. Accuracy and F1 are reported on MintVid (upper) and Average (lower). Quantity of “0” denotes the baseline.

a better foundation for detection. Interestingly, further combining A-G degrades the performance, which we suppose is due to the quality of A-G annotations. For A-G, we use data sampled from Molmo2 (Clark et al., 2026), which is coarse-grained (e.g., the entire person is masked out even only the arm deforms), making them less compatible with the fine-grained signals from SSL and G-G. We suppose the results might be different if *higher-quality artifact grounding annotations* are available. Overall, without any domain-specific annotations, using SSL and G-G as perceptual pretext tasks yields promising improvements. Detailed results can be found in Appendix Table 11.

Ablations on PPRL: the quantity of perception data. As shown in Figure 7, even introducing a small amount of perception data yields a measurable performance gain (i.e., +1.1% Acc and 1.3% F1 on MintVid with 1K perception data). As the amount increases, performance improves steadily. Excessive perception data does not further enhance detection accuracy, but it remains consistently better than the baseline. A scale around 5K is efficient and effective to deliver the best performance improvement.

Table 4. Difficulty of the SSL task. (.,.) denotes the range of values (“the number of pixels per side” for “Size” and “seconds” for “Duration”). “Baseline” is the model trained exclusively on AIGC detection task.

Difficulty	Size	Duration	ID		OOD		MintVid	
			Acc	F1	Acc	F1	Acc	F1
Baseline			92.9	93.4	91.2	92.1	77.8	73.3
Easy	(120, 240)	(3s, 5s)	92.5	92.9	91.3	92.2	79.5	76.3
Medium	(40, 120)	(2s, 4s)	92.9	93.1	92.5	93.3	79.6	76.0
Hard	(20, 40)	(1s, 3s)	93.1	93.5	92.3	93.4	80.9	77.3
Super-hard	(15, 20)	(0.2s, 1s)	92.7	93.2	91.3	92.4	80.8	77.1

Table 5. Attempts of batch-level perception integration. ‡ means removing the token normalization term when computing loss.

Method	ID		OOD		MintVid	
	Acc	F1	Acc	F1	Acc	F1
Baseline	92.9	93.4	91.2	92.1	77.8	73.3
Batch-level	84.6	86.4	88.4	88.8	76.8	74.1
Batch-level‡	89.7	90.7	91.4	92.1	79.2	76.4
Phase-level	93.1	93.5	92.3	93.4	80.9	77.3
Δ Batch-level‡	+3.4	+2.8	+0.9	+1.3	+1.7	+0.9

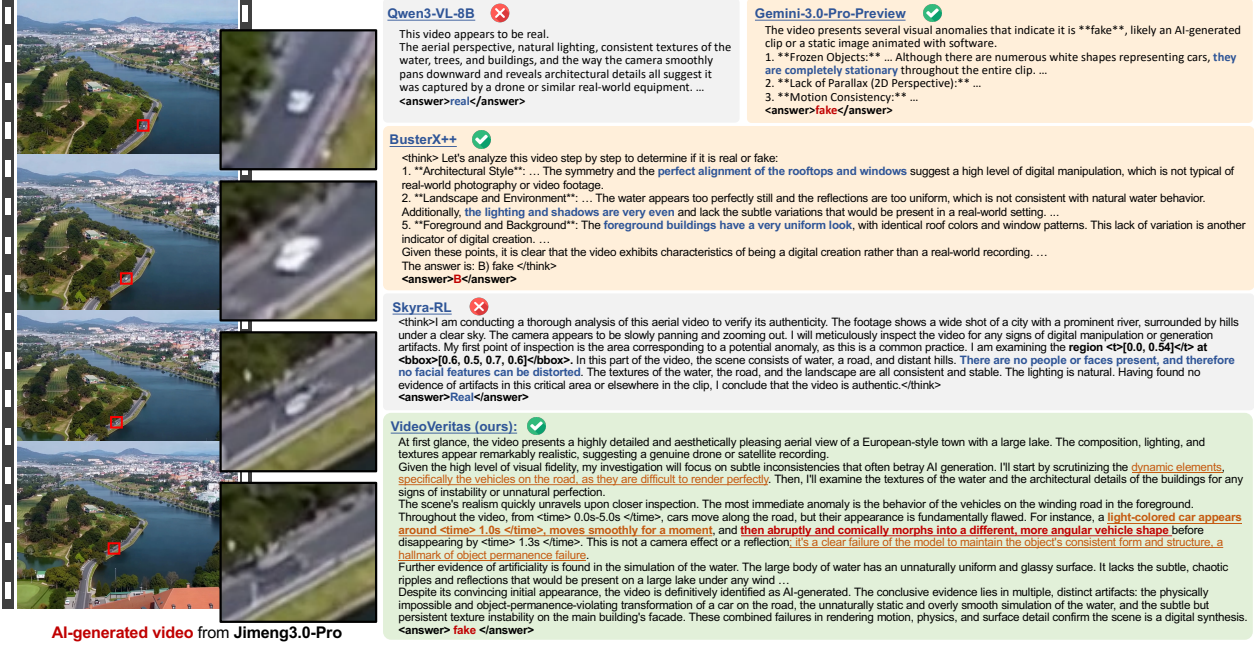


Figure 8. Comparisons between VIDEOVERITAS and existing MLLMs. A small car on the road distorts and disappears.

4.4. Further Analyses

Difficulty of perception task. As mentioned in Sec. 3.2, the difficulty of SSL task is controllable. We investigate different settings by adjusting the size and duration of the target objects, where smaller size and shorter exposure times make the task more challenging. As shown in Table 4, increasing the difficulty is beneficial to the detection performance, with the “Hard” setting achieving the best results. Actually, the “Hard” setting is already *non-trivial* for humans to perceive, yet the model can still be trained to count them accurately. This suggests that the perceptual capacity of *machines* may surpass that of *humans*, but remains underexploited.

Why could perception pretext task benefit detection. We provide more details about Figure 2. To facilitate *objective* evaluation, we (1) define five different dimensions and (2) perform pairwise evaluation rather than absolute scoring, where we randomly select 2K samples (1.5K fake and 0.5K real) that both the baseline and PPRL trained model yield correct answers. Gemini-3.0-Pro-Preview and GPT-5.1 are adopted for judgment and the results can be “win”, “lose” or “tie”. The results show that PPRL trained model tends to

analyze *fine-grained entities*, pay more attention to the *object relation* and performs better at *tracking changes*, which consequently benefits the detection. Evaluation prompts and the definition of the dimensions are listed in Appendix C.

Batch-level integration vs. Phase-level learning. Besides phase-level learning, we also explored batch integration, where perception and detection task are paired within each mini-batch and trained jointly. As shown in Table 5, it is non-trivial to directly perform batch integration. The outputs of detection task are significantly longer than perception counterparts, which limits effective gradients for detection under standard GSPO setting. We modified the loss computation to balance gradient allocation, which results in certain gains over baseline. However, phase-level learning remains more effective and is also simpler to apply in practice.

Case studies. As compared in Figure 8, the base model fails to capture fine-grained visual details. BusterX++ delivers correct answer, but the reasoning is superficial, focusing on elements like the overall context. Skyra-RL conducts template-like analysis and exhibits suboptimal perception on the high-quality video. In contrast, VIDEOVERITAS cor-

rectly identifies the distortion of the tiny car, which demonstrates superior fine-grained perception capacities.

5. Conclusion

In this paper, we introduced VIDEOVERITAS, a framework that integrates fine-grained perception and fact-based reasoning for AI-generated video detection. We introduce PPRL, which improves perceptual capacities by incorporating simple perception pretext tasks. We also introduce MintVid, a challenging dataset that contains three evaluation aspects. Extensive experiments demonstrate the effectiveness of VIDEOVERITAS, highlighting the value of learning foundational perceptual skills for complex detection tasks.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Acknowledgments

This work was supported in part by the New Generation Artificial Intelligence-National Science and Technology Major Project (No. 2025ZD0123501), the Beijing Natural Science Foundation JQ23016, the Chinese National Natural Science Foundation Projects 62476273 and 62406320, the Science and Technology Development Fund of Macau Project 0044/2024/AGJ, and 0084/2024/RIB2, and Ant Group Research Fund.

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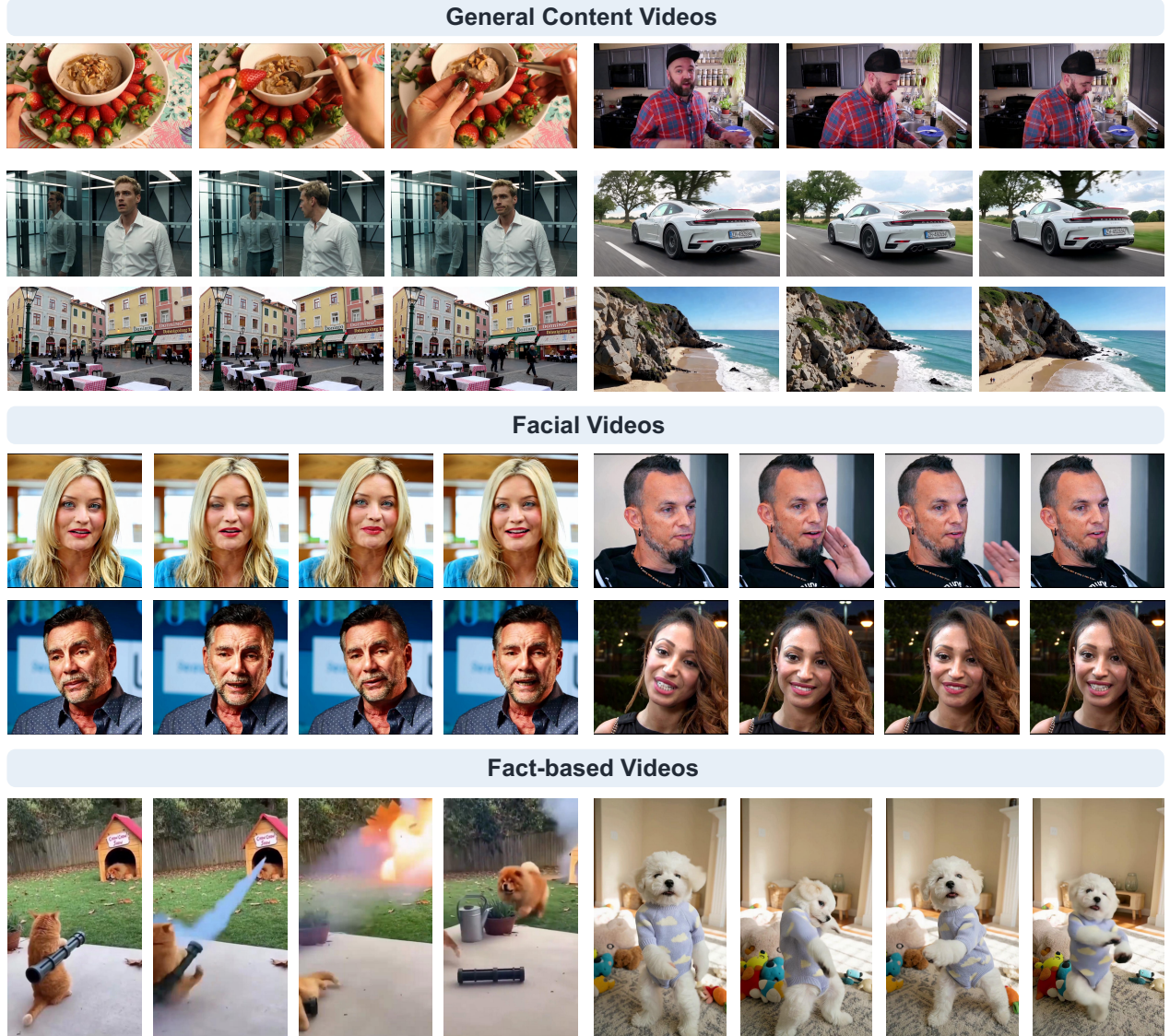


Figure 9. Visual examples of MintVid dataset.

A. More Details of MintVid Dataset

Generation pipeline of facial videos. To ensure the quality of the generated videos, we carefully curate the input prompts and conditional images. For the conditional images, we manually select frames from VFHQ (Xie et al., 2022) and HDTF (Zhang et al., 2021), ensuring the clarity of selected frames. For input prompts, we follow the commonly adopted structure, i.e., “**detailed caption of the first frame + action control + background description**”. Specifically, we use Qwen2.5-VL-32B (Bai et al., 2025b) to generate detailed caption for the input image. Then we curate about 20 action prompts, including control over facial expressions and gestures, e.g., “slowly turns head from left to right, then briefly dips chin downwards before returning to a neutral position”. We further curate about 20 background descriptions, covering different camera controls and environment settings, e.g., “The camera is fixed, creating a stable frame that emphasizes the subject’s presence. The lighting is soft and the background is tastefully blurred.”. Finally, we utilize Qwen2.5-VL-32B (Bai et al., 2025b) to check the plausibility of the prompts, e.g., whether the subject conflicts with the person’s gender in the image and whether background control contradicts the image content. An example of input prompts is provided in Figure 10.

Visual examples of MintVid dataset. The key attribute of MintVid dataset is that it considers three distinct aspects, aiming to facilitate more complete evaluation. As mentioned in Sec. 3.3, general content videos are generated using 6 proprietary models. Facial videos are generated using 3 specialized public models. Fact-based videos are collected from

An example of input prompt for facial video generation

A man is shown in a professional setting, likely within a governmental or official building. He is wearing a dark suit, a striped dress shirt, and a patterned tie. His hair is light-colored and neatly combed, and he is wearing rectangular-framed eyeglasses. The background features ornate columns and an American flag, indicating a significant or ceremonial context. He appears to be speaking or giving a statement, though his expression is calm and focused. He brings his left hand to his forehead as if in thought. The lighting mimics the soft, warm glow of the golden hour, casting a flattering light on the man. The ornate columns and American flag remain distinct in the background.

Figure 10. An example of input prompt for facial video generation.

online platforms and manually filtered. We provide more visual examples in Figure 9.

Summary of training data. To avoid ambiguity caused by rounded data counts, we summarize the training data used in different stages in Table 6. The J-DPO stage contains 4,219 samples, while the RL stage contains 10,789 AIGC samples and 5,000 perception samples, resulting in 20,008 training samples in total. The Pure RL baseline follows a comparable 20K training scale.

Table 6. Summary of training data used in different stages.

Stage	Data Type	Source	Samples
J-DPO	AIGC detection videos	GenBuster-200K, RewardData, TalkingheadBench, OpenVid, and fact-based videos from YouTube	4,219
RL (perception stage)	Perception pretext data	Grounding data from OneThinker and counting data constructed from OpenVid	5,000
RL (detection stage)	AIGC detection videos	Same sources as J-DPO	10,789
Total	-	-	20,008
Testing Data	-	GenBuster++, LOKI, VBench-2, MVAD, MintVid	-

B. Artifacts Taxonomy

To enable more fine-grained analyses, we summarize a taxonomy of common artifacts observed in AI-generated videos, covering 3 main perspectives and 11 detailed aspects.

Motion-level Anomalies:

- **Unnatural Kinematics and Trajectories:** The movement of an object or being defies physical dynamics or biomechanics in its speed, acceleration, or path.
- **Object Permanence Failure:** An object’s existence or individuality is violated, causing it to suddenly appear from nowhere, abruptly vanish without reason, illicitly merge with other distinct entities (e.g., two people blending into a single mass), or spontaneously split into multiple copies.
- **Structural Integrity Failure:** A single object, while maintaining its presence, fails to preserve its own physical form, causing its structure or identifying details to illogically warp, stretch, tear, or distort during motion. The object behaves as if made of the wrong material (e.g., a rigid pole bending like rubber).
- **Interaction Anomalies:** Objects violate their physical properties upon contact, or the scene’s spatial coherence breaks down during camera movement.
- **Biological Motion Irregularity:** The natural, subtle dynamic features of living beings, such as blinking, breathing, or micro-expressions, are absent, stiff, or unnaturally timed.

Physical-level Anomalies:

- **Inconsistent Lighting and Optics:** Shadows, highlights, or reflections behave inconsistently with the scene’s established light sources, object geometry, or movement.

- **Causality and Property Violation:** The sequence of events defies cause-and-effect logic, or an object’s behavior contradicts its inherent physical properties.
- **Flawed Material Simulation:** The dynamic simulation of complex, non-rigid materials like cloth, smoke, fire, or water appears unrealistic and lacks natural behavior.
- **Contextual and Semantic Mismatch:** The combination of scenes, objects, and actions is logically or commonsensically contradictory, even if individual elements are rendered realistically.

Perceptual-level Anomalies:

- **Texture and Surface Instability:** An object’s surface texture exhibits a high-frequency flicker, “crawling” pattern, or shimmer that is disconnected from its motion.
- **Definition and Clarity Fluctuation:** The sharpness and detail level of a region abruptly and illogically degrade or change without a corresponding cause.

C. Extended Details of Methodology: Prompts and Implementation Details

Details of reasoning behavior analysis (Figure 2). We adopt 5 objective dimensions to investigate the model’s reasoning behavior:

- **Component Granularity:** This dimension evaluates whether the assistant describes the scene as a whole or deconstructs it into specific objects and their fine-grained components (e.g., specific limbs, lens edges, or individual textures).
- **Spatiotemporal Continuity:** This dimension evaluates the assistant’s ability to anchor its observations to precise temporal markers (timestamps) and spatial locations. It looks for how well the analysis “tracks” changes over a specific timeline.
- **Physics Depth:** This dimension evaluates the extent to which the assistant uses principles of physics (optics, mechanics, biology) to explain anomalies, rather than simply stating that something “looks wrong”.
- **Forensic Objectivity:** This dimension evaluates the shift from subjective, impression-based language (e.g., “uncanny”, “beautiful”) to objective, evidence-based descriptions (e.g., “static texture overlay”, “non-uniform deformation”).
- **Relational Logic:** This dimension assesses the analysis of how different elements in the report interact with one another (e.g., the relationship between a moving object and its shadow, or the reaction of a surface to a force).

The evaluated samples are correctly answered by both models, which avoids the answer bias. For each dimension, we compare the outputs of baseline (trained without perception data) and our model (trained with PPRL). To mitigate potential bias, we utilize two powerful model, i.e., Gemini-3.0-Pro-Preview and GPT-5.1 to conduct pairwise comparison. The input prompts are presented in Figure 19. Specifically, the judging models are required to provide a brief analysis and judgment: model A is better, model B is better or the two models are tied. Then, we adopt the win rate as a straightforward and unbiased evaluation of the performance. An example of the judgment is shown below.

Prompts of VIDEOVERITAS. Our framework involves perception tasks and detection task, and we do not require careful curation of input prompts. For detection, the system prompt and input prompt are presented in Figure 12 and Figure 13. For perception, the system prompt is shared as presented in Figure 14. The input instructions are different among different sub-tasks, as shown in Figure 15, Figure 16, Figure 17 and Figure 18.

More implementation details. Our model is trained with 8 PPUE GPUs based on ms-swift (Zhao et al., 2024). The theoretical peak computational capacity (TFLOPS) of one PPUE GPU is roughly half of an NVIDIA A100 GPU, and each PPUE GPU has 96GB VRAM. Suppose the training dataset of RL stage is denoted as $\mathcal{D}$, the objective function of GSPO is formulated as:

$$\mathcal{L}_{\text{GSPO}} = -\mathbb{E}_{(v, y) \sim \mathcal{D}, \{\mathbf{o}_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|v)} \frac{1}{G} \sum_{i=1}^G \frac{1}{|\mathbf{o}_i|} \sum_{t=1}^{|\mathbf{o}_i|} [\min(r_i(\theta)A_{i,t}, \text{clip}(r_i(\theta), 1-\epsilon_1, 1+\epsilon_2)A_{i,t})], \quad (9)$$

Judgment Output of “Component Granularity”

“analysis”: “Assistant A demonstrates superior granular analysis by deconstructing the scene into specific components with detailed observations. **It isolates and analyzes specific elements** such as “**small bubbles**” (noting their spherical shape and static nature), distinct light reflections (described as solid, glowing shapes), and the specific “sun” object (a glaring red dot) versus the color of the light on the water. **In contrast, Assistant B focuses on broader features** like the general “oil slick”, “edges”, and “overall scene”, lacking the fine-grained decomposition found in A.”,
 “judgment”: “[A]”

Figure 11. Example of judgement output to “Component Granularity”. The judging model is Gemini-3.0-Pro-Preview.

where importance ratio r_i is defined based on sequence likelihood:

$$r_i(\theta) = \left(\frac{\pi_\theta(\mathbf{o}_i|\mathbf{v})}{\pi_{\theta_{\text{old}}}(\mathbf{o}_i|\mathbf{v})} \right)^{\frac{1}{|\mathbf{o}_i|}}, \quad (10)$$

and the advantage is calculated based on group estimation:

$$A_{i,t} = \frac{R_i - \text{mean}(\{R_1, \dots, R_G\})}{\text{std}(\{R_1, \dots, R_G\})}. \quad (11)$$

Specifically, G is set to 4 in our method. ϵ_1 and ϵ_2 are set to 3×10^{-4} and 4×10^{-4} , respectively. In the ablations of Table 5, we found that it is non-trivial to perform batch-level perception integration with the above formulation, since the objective is normalized by the output length in Eq. 9. To balance the gradient between different tasks, we remove the normalization term $\frac{1}{|\mathbf{o}_i|}$ in Eq. 9. This greatly elevates the performance, which is superior than the baseline, demonstrating that incorporating perception learning is beneficial to our detection task.

D. More Experimental Results

More metrics. As presented in Table 10, we further report recall and F1 (with “fake” taken as positive category). Specifically, NSG-VD exhibits high recall on several subsets, while the overall accuracy is near-chance level. RestraV suffers from low recall on most datasets. Zero-shot MLLMs tend to predict videos as real, exhibiting low recall on most datasets. However, Gemini-2.5-Pro and Gemini-3-Pro-Preview show *clear and decisive leads* in recall, indicating that their perceptual capabilities remain significantly stronger than those of open-source models. Notably, most models achieve near-perfect results on Emu3, which is collected from MVAD (Hu et al., 2025). The reason may come from two aspects: (1) most videos in Emu3 are in an anime style, which are generally easy for MLLMs to judge, and (2) the distribution produced by autoregressive generators like Emu3 may be quite different. BusterX++ performs well on most datasets, but exhibits obvious shortcomings on datasets like OmniAvatar and FantasyTalking. Skyra-RL fails on most datasets, achieving only 6.9% averaged recall, which as we mentioned in the main text, may be due to its controlled training procedure. Overall, VIDEOVERITAS yields more balanced performance, achieving superior recall and F1 score over previous methods.

More comparisons. We provide more comparisons of reasoning outputs in Figure 21, Figure 22, Figure 23, Figure 24 and Figure 25. Although Skyra-RL can conduct grounded analysis by examining specific regions, their analyses are mechanical and imprecise, which leads to failures on those fact-based videos, e.g., AI parodies in Figure 25. BusterX++ gets correct answers on most samples, but the *reasoning content is superficial*, mainly focus on the macro-level semantic concepts like clothing and lighting. In contrast, VIDEOVERITAS demonstrates superior fine-grained perceptual capacity and fact-based reasoning ability. For instance, in Figure 21 upper, previous methods all fail to perceive the distortion of the person’s feet when entering the boat, while VIDEOVERITAS precisely captures this artifact. In Figure 21 lower, VIDEOVERITAS correctly captures the object permanence failure, where a bottle emerges suddenly and the materials in the bottle have unexpectedly changed.

Perception capacity of strong MLLMs. A common concern is that those proprietary MLLMs had been trained on extensive perception data, why could thousands of samples in our PPRL bring improvements? The key observation is that these powerful models still fall short on highly difficult samples. As shown in Table 7, we present the performance on our

sampled tracking and counting tasks. Gemini-3.0-Pro-Preview struggles under challenging settings such as Counting-Hard. Meanwhile, open-source MLLMs still lag behind proprietary models on perception-heavy tasks. These results suggest that large-scale generic pretraining remains insufficient for granular spatiotemporal perception, while PPRL provides targeted signals for hard, localized, and temporally precise observations.

Reasoning-quality evaluation. As shown in Table 8, we conduct both MLLM-as-a-Judge and human judgment to evaluate the quality of model explanations. For MLLM-as-a-Judge, we randomly sample 1,000 videos and their reasoning outputs (800 fake and 200 real), and use Gemini-3.0-Pro-Preview for pairwise comparison. For human judgment, we randomly sample 200 videos (150 fake and 50 real), forming 600 pairwise comparisons among three models. Annotators are asked to compare explanations from three perspectives: perception accuracy, forensic objectivity, and logical coherence. Following prior practice, we compute the final ranking with Elo rating. The results show that **VIDEOVERITAS** achieves the highest Elo score under both automatic and human protocols, which is consistent with the qualitative comparisons. The human-judgment platform screenshot is provided in Figure 20.

Performance of image-level detection. Although **VIDEOVERITAS** is trained exclusively on video data, its core capacities, i.e., fine-grained perception and fact-based reasoning, are also useful for image-level detection. As shown in Table 9, **VIDEOVERITAS** achieves competitive results on both GenBuster++ (Wen et al., 2025b) and Chameleon (Yan et al., 2024a) benchmarks, indicating that the learned perceptual and reasoning skills can transfer beyond the video setting.

Brief summary of explored attempts. We briefly summarize several unsuccessful attempts. (1) For batch-level integration, we explored conditional reward designs for tight coupling, but it did not achieve improvements comparable to phase-level learning. (2) We tried more forms of SSL like temporal anomaly grounding (e.g., generated by shuffling local clip), but they were less effective than spatiotemporal tasks like object counting. (3) We also tried more detailed reward designs for detection task, e.g., explicitly encouraging grounding formats like `<time> </time>`, which were found to be vulnerable to reward hacking. However, we suppose the results drawn from such heuristic designs may *differ* when high-quality annotations are available.

Table 7. Perception and detection performance of strong MLLMs. Counting, Tracking, and MintVid accuracy are reported. “M” stands for Medium, and “H” represents Hard.

Model	Count-M	Count-H	Tracking	MintVid
Gemini-3.0-Pro-Preview	79.4	50.8	42.4	72.1
Qwen3-VL-235B-A22B	79.5	25.5	36.5	56.1
GLM-4.5V	75.1	34.7	39.8	56.6
VIDEOVERITAS (8B)	93.5	72.9	48.3	80.9

Table 8. Elo scores for reasoning-quality evaluation. Higher scores indicate better explanation quality. Gemini-3.0-Pro-Preview is adopted for MLLM-as-a-Judge.

Method	MLLM-as-a-Judge	Human-Judge
Skyra	784.30	724.50
BusterX++	963.29	1089.21
VIDEOVERITAS	1252.41	1186.30

Table 9. Transferability to image-level detection. The reported metrics are “Acc/Recall/F1”.

Method	Training	GenBuster++	Chameleon
Qwen3-VL-8B	–	62.1/28.9/43.4	72.0/57.2/63.7
FakeVLM	Image	55.5/ 86.5 /66.0	62.9/ 99.7 /69.7
BusterX++	Image+Video	78.1/75.7/77.6	76.1/80.5/74.3
VIDEOVERITAS	Video	86.4 /76.2/ 84.9	79.8 /88.9/ 79.1

E. Limitation and Future Work

One limitation of our work is that the perception pretext data used in PPRL mainly comes from broadly available public resources (e.g., generic grounding/tracking and self-supervised counting). While this choice keeps the training pipeline lightweight and scalable, we do not systematically investigate whether higher-quality, artifact-specific grounding annotations could further improve detection performance and explanation fidelity. Exploring and curating such high-quality artifact grounding data, as well as integrating it into the pretext RL stage, is a promising direction for future work.

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Table 10. Performance comparison on video datasets, including Accuracy (Acc), Recall (R) and F1. * means Average Precision (AP) is adopted following the official guideline. The final average performance is calculated by first averaging the results across ID (in-domain), OOD (out-of-domain) and OOD-MintVid, and then taking the average of the three values. The best results are **bolded**.

Method	Metric	ID	OOD							OOD-MintVid										Avg.
			GenBust++	LOKI	Vidu Q1	Gen-4	Veo3	Emu3	Phantom-14B	OmniAvatar	FantasyTalking	Seedance1.0Pro	Jimeng3.0Pro	Kling2.5-Turbo	Hailuo2.3	Wan2.5	Sora2	Fact		
Small Vision Models																				
D3* (ICCV'25)	AP	49.5	46.8	39.1	55.1	89.0	66.1	90.0	54.8	36.4	44.1	61.6	48.7	51.7	78.9	50.5	54.6	32.4	55.1	
	Acc	87.6	86.8	71.1	80.7	83.4	79.4	85.9	57.4	55.2	62.2	49.2	46.2	46.6	41.1	49.3	41.4	73.3	73.7	
	R	87.2	90.8	84.4	77.7	81.5	77.7	87.8	50.4	58.1	60.0	72.4	57.6	58.3	54.1	64.1	48.1	73.8	76.7	
DeMamba	F1	87.3	87.3	79.5	80.1	83.1	79.1	86.1	54.2	57.9	61.3	51.1	50.6	51.2	35.1	54.8	43.0	79.5	74.6	
	Acc	52.4	53.3	55.1	53.4	58.2	64.3	50.7	54.3	53.8	58.5	53.8	58.0	56.0	69.4	57.0	57.8	52.8	55.1	
	R	28.0	76.0	75.2	60.2	84.4	87.2	1.5	48.2	86.4	67.5	17.2	36.1	75.9	83.1	80.9	77.3	30.6	50.8	
NSG-VD (NIPS'25)	F1	27.9	61.9	62.5	56.4	66.9	71.0	3.0	51.4	65.1	61.9	27.1	46.3	63.3	73.2	65.3	64.8	39.3	45.8	
	Acc	50.7	48.9	64.8	54.8	52.8	59.8	95.2	49.6	49.3	49.9	36.9	48.4	47.2	30.2	40.9	45.4	65.0	53.2	
	R	4.3	3.8	5.6	7.8	8.0	18.2	96.2	3.7	3.7	3.7	2.5	2.5	2.5	2.5	2.5	2.5	11.1	10.4	
ReStraV (NIPS'25)	F1	8.0	6.9	9.7	14.0	14.5	30.0	95.3	6.8	6.8	6.8	4.9	4.9	4.8	4.9	4.8	4.8	16.0	14.3	
	Generic MLLMs																			
	Qwen2.5-VL-7B	Acc	54.1	59.5	49.4	58.8	53.3	60.3	99.2	50.7	50.0	49.7	63.6	49.4	51.6	53.1	55.8	50.4	87.1	57.9
R		13.5	23.5	27.9	15.4	8.0	18.2	99.7	1.2	0.1	0.0	14.7	6.3	7.7	8.8	19.0	6.2	84.6	20.1	
F1		21.7	36.2	42.2	26.1	14.7	30.2	99.3	2.3	0.3	0.0	23.9	11.3	13.7	15.6	30.7	11.2	87.8	27.6	
Qwen3-VL-8B	Acc	65.1	62.4	56.6	60.2	59.4	62.9	99.9	53.3	50.6	50.3	64.2	51.8	55.1	54.1	66.7	50.8	94.6	63.6	
	R	33.1	28.5	38.8	20.5	18.8	26.3	100.0	7.0	1.4	0.9	20.7	12.5	19.2	13.0	41.6	7.4	93.6	31.2	
	F1	44.3	43.1	54.2	34.0	31.6	41.5	99.9	13.0	2.8	1.7	31.8	21.1	30.7	21.8	56.4	13.0	92.6	41.2	
InternVL3.5-8B	Acc	51.7	59.6	44.1	56.1	53.4	56.7	99.7	51.3	50.2	49.5	62.6	49.8	49.1	50.2	53.9	54.1	91.7	56.5	
	R	9.3	8.6	14.2	9.7	6.3	5.9	99.5	1.3	1.9	0.9	5.9	3.2	4.1	2.2	7.8	5.3	85.1	15.0	
	F1	16.3	15.5	24.5	17.3	11.9	11.2	99.7	2.5	3.8	1.8	10.9	6.2	7.8	4.3	14.2	10.0	91.4	20.5	
MiMo-VL-7B	Acc	65.8	54.3	53.6	65.5	65.8	66.0	98.8	58.4	50.1	50.6	69.7	56.7	61.7	56.1	66.0	55.4	89.9	64.8	
	R	42.4	20.3	37.0	32.6	33.2	34.0	100.0	21.8	5.2	6.2	33.2	21.5	31.5	16.4	39.6	16.2	95.0	38.0	
	F1	52.7	30.7	51.4	48.5	49.3	50.0	98.8	34.4	9.5	11.2	46.9	33.9	46.0	27.0	54.7	26.5	91.1	48.5	
Keye-VL-1.5-8B	Acc	52.8	62.4	60.4	58.3	56.0	62.4	94.7	51.3	51.1	53.3	59.7	50.5	51.0	52.8	52.9	51.8	80.4	58.0	
	R	13.3	29.0	39.2	14.4	14.8	29.0	97.7	3.9	4.6	7.4	23.9	13.1	13.4	13.5	18.1	16.0	84.0	23.5	
	F1	19.9	43.8	50.4	24.7	24.6	43.8	94.0	7.3	8.7	13.6	34.7	21.2	21.5	22.2	28.0	25.2	81.3	31.1	
GLM-4.5V	Acc	61.7	59.4	49.4	56.5	56.6	62.4	97.1	57.9	52.2	50.5	63.2	48.9	49.2	49.8	58.3	47.9	88.2	60.6	
	R	30.4	27.7	32.0	26.6	17.3	29.8	99.2	18.4	6.5	2.6	27.4	16.6	15.7	13.8	32.7	11.9	95.2	31.1	
	F1	43.3	40.6	46.4	40.4	28.5	44.2	97.1	30.3	12.0	5.0	39.2	26.8	25.5	22.9	46.7	19.9	90.1	41.6	
Qwen3-VL-235B-A22B	Acc	66.7	64.6	57.9	57.4	65.7	63.4	98.5	54.1	50.9	51.2	65.8	49.1	32.3	49.8	63.5	50.5	93.6	63.6	
	R	38.7	36.5	44.7	29.3	31.9	28.7	100.0	9.1	2.6	3.3	32.3	16.2	32.3	13.7	41.3	16.2	94.3	36.7	
	F1	51.6	50.8	59.4	45.0	48.1	43.8	98.5	16.5	4.9	6.3	45.1	26.4	48.8	22.7	55.8	26.2	94.3	48.0	
Gemini-2.5-Pro	Acc	80.5	71.8	70.2	70.8	76.0	70.4	93.3	57.8	55.8	55.6	69.4	60.2	65.6	62.7	65.7	60.9	88.3	73.4	
	R	85.1	59.8	65.5	38.2	65.6	46.8	100.0	28.2	22.6	18.2	52.0	34.9	47.7	39.4	47.7	39.4	95.5	63.4	
	F1	81.4	68.1	70.3	55.3	74.7	60.4	93.7	40.6	33.8	27.9	60.9	45.9	58.5	51.0	58.4	51.0	90.3	67.9	
Gemini-3-Pro-Preview	Acc	79.4	82.5	74.9	83.1	76.3	85.7	89.3	65.5	65.3	61.5	81.4	72.1	73.1	74.3	73.1	63.5	91.4	77.8	
	R	96.2	93.5	94.4	92.7	75.8	89.6	100.0	60.8	60.2	52.7	81.2	71.8	73.9	76.0	73.9	54.4	98.6	86.3	
	F1	82.7	84.3	80.2	84.7	76.1	86.3	90.3	63.9	63.4	57.6	81.3	71.9	73.1	74.5	73.3	59.7	92.6	79.1	
MLLM-based Video Forgery Detectors																				
BusterX++	Acc	77.1	79.6	70.9	53.0	89.9	76.2	98.9	62.3	48.6	47.6	71.8	71.7	82.3	70.4	87.3	58.7	88.8	74.7	
	R	64.9	68.8	67.5	6.1	82.9	55.5	100.0	30.6	3.3	1.3	47.8	47.3	73.9	45.6	86.2	13.0	90.0	57.4	
	F1	71.4	77.1	75.5	11.4	89.2	70.0	98.9	44.8	6.0	2.4	57.7	57.2	77.1	55.6	84.5	19.8	89.7	63.7	
Skyra-RL	Acc	52.1	55.7	37.7	52.9	59.1	55.9	51.5	50.1	49.9	49.9	63.4	51.7	53.5	51.7	59.6	50.6	51.9	52.5	
	R	4.6	11.7	7.3	6.1	18.2	12.1	3.0	0.3	0.0	0.0	9.9	6.9	10.8	2.7	22.3	1.0	11.4	6.9	
	F1	8.4	20.9	13.5	11.4	30.7	21.5	5.8	0.6	0.0	0.0	17.9	12.8	19.4	5.3	36.4	2.0	20.5	12.4	
VIDEOVERITAS (ours)	Acc	93.1	91.4	78.1	93.6	96.7	94.5	99.4	79.0	56.2	84.5	86.6	80.8	86.0	85.8	86.3	67.6	96.1	88.8	
	R	99.3	94.9	91.0	89.5	96.0	92.3	100.0	76.6	30.5	87.5	75.7	68.4	78.6	78.1	79.3	41.1	100.0	88.3	
	F1	93.5	91.7	84.7	93.3	96.7	94.3	99.4	78.5	40.9	84.9	82.0	78.6	85.4	84.4	85.8	55.8	96.6	88.0	

Table 11. Detailed ablation of perception tasks on MindVid dataset. The reported metrics are “Acc/Recall/F1”.

Task	Phantom-14B	OmniAvatar	FantasyTalking	Seedance 1.0Pro	Jimeng3.0Pro	Kling2.5-Turbo	Hailuo2.3	Wan2.5	Sora2
w/o Perception	77.1/76.0/76.8	52.6/27.1/36.4	84.7/91.2/85.6	81.3/67.5/74.5	76.2/62.6/73.1	83.5/76.9/82.9	82.8/75.3/81.2	83.0/75.9/82.2	61.1/31.3/44.5
SSL	81.0/86.1/81.9	62.5/49.1/56.7	85.5/95.2/86.8	84.1/74.3/78.9	78.9/67.9/76.9	83.7/77.2/83.1	82.4/78.1/81.4	84.6/78.9/84.1	65.5/40.1/53.6
G-G	76.2/72.3/75.3	58.2/36.2/46.4	84.2/88.2/84.8	85.3/76.4/80.7	79.4/68.2/77.4	85.1/79.3/84.7	86.1/82.9/85.5	83.9/77.1/83.3	66.5/41.4/55.2
A-G	79.5/79.7/79.5	58.2/37.2/47.1	85.2/91.2/86.1	84.5/73.9/79.4	79.4/67.8/77.2	84.3/77.5/83.7	86.1/82.2/85.4	84.4/77.6/83.8	68.7/45.5/59.1
SSL + G-G	79.0/76.6/78.5	56.0/30.5/40.9	84.5/87.5/84.9	86.6/75.7/82.0	80.8/68.4/78.6	86.0/78.6/85.4	85.8/78.1/84.4	86.3/79.3/85.8	67.6/41.1/55.8
SSL + A-G	79.8/80.6/79.9	58.8/38.6/48.4	86.2/93.3/87.1	82.7/70.0/76.5	78.1/65.7/75.6	83.8/76.9/83.1	83.8/78.1/82.6	83.6/76.5/82.9	64.2/36.7/50.5
G-G + A-G	78.1/80.4/78.6	57.1/38.4/47.3	84.9/94.0/86.2	86.4/79.6/82.5	79.7/69.2/77.9	85.1/79.6/84.7	85.1/81.5/84.4	84.8/78.9/84.3	63.7/36.0/49.7
SSL + G-G + A-G	79.1/76.9/78.7	60.1/38.9/49.4	87.8/94.2/88.5	81.3/68.5/74.7	74.4/59.8/70.7	81.9/74.4/81.0	81.1/75.3/79.7	83.3/77.1/82.7	64.2/38.0/51.4

System Prompt for Detection

You are an expert video analyst.
Please think about the question as if you were a human pondering deeply. It's encouraged to include self-reflection or verification in the reasoning process. Then, give a final verdict within `<answer>` `</answer>` tags.

Figure 12. System prompt for detection task.

User Prompt for Detection

`<video>`
Is this video real or fake?

Figure 13. User prompt for detection task.

System Prompt for Perception

You are an expert video analyst. Based on this video, provide the answer directly.

Figure 14. System prompt for perception task.

User Prompt for Spatiotemporal Grounding

`<video>`
Given the query “[description]”, when and where does the described content occur in the video? please firstly give the start and end time, spatial bounding box corresponding to each integer second.

Please provide only the time span in seconds and bounding boxes as JSON, ONLY up to 16 seconds. You MUST output one bounding box for every integer second within the given time span (inclusive).

Example:
{“time”: [8.125, 13.483], “boxes”: {“9”: [317, 422, 582, 997], “10”: [332, 175, 442, 369], “11”: [340, 180, 450, 370]}}

Note: Each key in “boxes” must be an integer second within the span, and its value must be a 4-number bounding box [x1, y1, x2, y2].

Figure 15. User prompt for spatiotemporal grounding task. “[description]” is a brief caption of the target object, which is included in the dataset, such as “a baby closes a laptop in the study”.

User Prompt for Object Tracking

`<video>`
Given the bounding box “[initial box]” of the target object in the first frame, track this object in each frame and output its bounding box once per second. ONLY up to 16 seconds.

Example:
{“boxes”: {“1”: [405, 230, 654, 463], “2”: [435, 223, 678, 446], ..., “16”: [415, 203, 691, 487]}}

Note: Each key in “boxes” must correspond to a second (1, 2, 3, ..., 16) and contain a 4-number bounding box [x1, y1, x2, y2].

Figure 16. User prompt for object tracking task. “[initial box]” is the bounding box in the first frame, which is included in the dataset.

User Prompt for Object Counting

<video>
Count the number of circles, squares, and triangles that appear in this video. Be aware that the shapes can appear in any color and at any angle of rotation. They may be present on one or multiple frames, and any given frame can contain more than one shape. Provide the answer as three comma-separated numbers in the format: circles,squares,triangles. For example, if you see 3 circles, 1 square, and 4 triangles, your answer should be “3,1,4”.

Figure 17. User prompt for object counting task.

User Prompt for Artifact Grounding

<video>
Find the visual artifacts at “[time]” in the video.
Provide the bounding boxes where the artifact occurred, in $[x_{min}, y_{min}, x_{max}, y_{max}]$ format. If there are multiple locations, you should provide all the bounding boxes.
Example:
{“boxes”: [[487, 324, 573, 398], [670, 533, 734, 769], ...]}.

Figure 18. User prompt for artifact grounding task (as mentioned in Sec. 4.3). “[time]” is the exact target timestamp in seconds, which is included in the dataset, such as “4.5s”.

Input Prompt for Reasoning Behavior Evaluation

You are a helpful assistant proficient in analyzing vision reasoning problems.
Instruction:
You will be presented with two analytical reports (Assistant A and Assistant B) that describe observations from a video. Your task is to perform a side-by-side comparison and determine which assistant provides higher-quality reasoning based **ONLY** on the specific dimension provided below.
The evaluation must be conducted strictly based on the textual evidence provided. Do not assume any external video information. Your goal is to identify which assistant demonstrates more professional, precise, and logically structured analysis within the specified scope.

Evaluation Dimension: {Dimension Name}
Description: {Dimension Description}

Rules for Evaluation:

- **Strictly Dimension-Focused:** Ignore other aspects of the reports. Only judge based on the provided dimension.
- **Content over Conclusion:** Do not favor an assistant based on its final verdict. Focus on the depth and quality of the reasoning path.
- **Neutrality:** Be unbiased toward length or tone. Prioritize the density of meaningful, analytical information.

Desired Output Format:
Present your verdict in a JSON format, with key “analysis” for a short reason of your judgment and key “judgment” to indicate your decision: use “[A]” if assistant A prevails, “[B]” if assistant B does, and “[C]” for a tie.

Input Data:
[The Start of Assistant A’s Analysis]
{Output of Assistant A}
[The End of Assistant A’s Analysis]
[The Start of Assistant B’s Analysis]
{Output of Assistant B}
[The End of Assistant B’s Analysis]

Figure 19. Input Prompt for Reasoning Behavior Evaluation (Figure 2).



Figure 20. Human-judgment platform for pairwise reasoning-quality comparison.



Figure 21. Reasoning comparisons between VIDEOVERITAS and existing MLLM-based detectors.



Figure 22. Reasoning comparisons between VIDEOVERITAS and existing MLLM-based detectors.



Figure 23. Reasoning comparisons between VIDEOVERITAS and existing MLLM-based detectors.



Figure 24. Reasoning comparisons between VIDEOVERITAS and existing MLLM-based detectors on facial videos.

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Figure 25. Reasoning comparisons between VIDEOVERITAS and existing MLLM-based detectors on fact-based videos.